Geospatial File Measurement API

Problem Statement

Build a backend service using **either Django or FastAPI** that accepts a geospatial file, processes the features within it, and returns measurement information.

The service should accept either:

- **Shapefile**
- **KML**

The objective is to evaluate your ability to build a clean backend API, work with geospatial data, handle file processing, and write production-quality Python code.

You are free to make architectural and implementation decisions beyond the minimum requirements.

Minimum Requirements

1. Backend Framework

Build the application using **one** of:

- Django + Django REST Framework
- FastAPI

2. File Upload

Provide an API endpoint that accepts a geospatial file.

Example:

POST /api/files/

The API should accept:

- .zip containing a Shapefile

- .kml

3. File Processing

After uploading the file, the service should read the geospatial data and extract its features.

For every feature, identify at minimum:

- Feature ID/index
- Geometry type
- Geometry
- CRS
- Properties/attributes

If a geometry type is not supported for measurement calculation, the API should handle it gracefully rather than crashing.

4. Measurements

Calculate measurements for supported geometries.

Polygon

Calculate:

- Area

LineString

Calculate:

- Length

Point

No measurement is required.

5. CRS Handling

The application must correctly handle files with geographic coordinates.

For example, a file may use:

EPSG:4326

The application should **not directly calculate area or distance using latitude/longitude degrees**.

Transform the geometry to an appropriate projected coordinate system before calculating measurements.

You may choose your own strategy for selecting the projected CRS.

6. API Design

At minimum, provide APIs for:

Upload

POST /api/files/

Uploads and processes a file.

File Information

GET /api/files/{id}/

Returns information about the uploaded file.

Example:

```
{
  "id": "abc123",
  "filename": "survey.kml",
  "feature_count": 120,
  "crs": "EPSG:4326",
  "status": "COMPLETED"
}
```

Measurements

GET /api/files/{id}/measurements/

Returns measurements for the features in the file.

7. Documentation

Include a README.md containing:

Setup

Explain how to run the application locally.

API

Document the available endpoints and example requests/responses.

Architecture

Briefly explain:

- Application structure
- File-processing flow
- Measurement calculation flow
- CRS handling

Design Decisions

Explain important technical decisions and alternatives you considered.

8. Submission

- Create a public github repo and share the link.
- In the readme mention learning and future scope.